Optimizing Optical Inter-Satellite Links for Non-Terrestrial Networks

A **strategic framework** for balancing laser **physics, pointing alignment,** and network latency in mega-**conste**llations.

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The Paradigm Shift to Optical Link Architecture

Incumbent: Radio Frequency (RF)

Spectrum Bandwidth:
- Congested S, Ka, Ku bands.

Hardware Footprint:
- Large antennas, high power requirements.

Signal Propagation:
- Omnidirectional, high path loss.

Primary Impairment:
- Spectrum congestion, interference.

The Future: Optical (OISL)

Spectrum Bandwidth:
- Massive 300 GHz to 430 THz range.

Hardware Footprint:
- Reduced Size, Weight, and Power (SWaP).

Signal Propagation:
- Highly concentrated, targeted energy.

Primary Impairment:
- Pointing errors, microscopic jitter.

The Vulnerability of the Vacuum: While space eliminates atmospheric turbulence, it introduces a critical bottleneck. Microscopic satellite vibrations completely disrupt high-frequency laser alignment over vast distances.

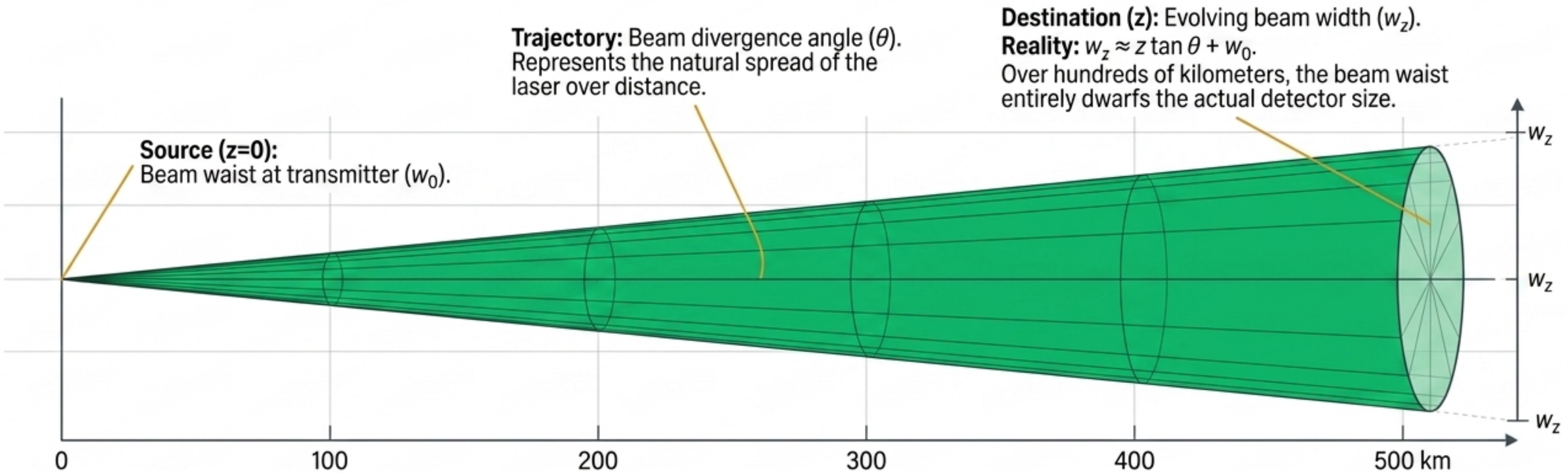
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Bridging the Diagnostic Gap in Laser Propagation

✗ **Omnidirectional Assumption:** Prior models treat space signals like terrestrial RF, ignoring directional focus.	✓ **Gaussian Beam Modeling:** We strictly account for highly directional, concentrated photon energy.
✗ **Fixed Beam Waist:** Literature ignores energy diffusion over long distances.	✓ **Evolving Beam Width:** We model dynamic beam waist expansion across thousands of kilometers.
✗ **Atmospheric Focus:** Existing optical models prioritize terrestrial air turbulence.	✓ **Vacuum-Specific Modeling:** We isolate satellite jitter, tracking noise, and Rayleigh-distributed radial deviation as the primary failure modes.

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Anatomy of an Expanding Laser Link

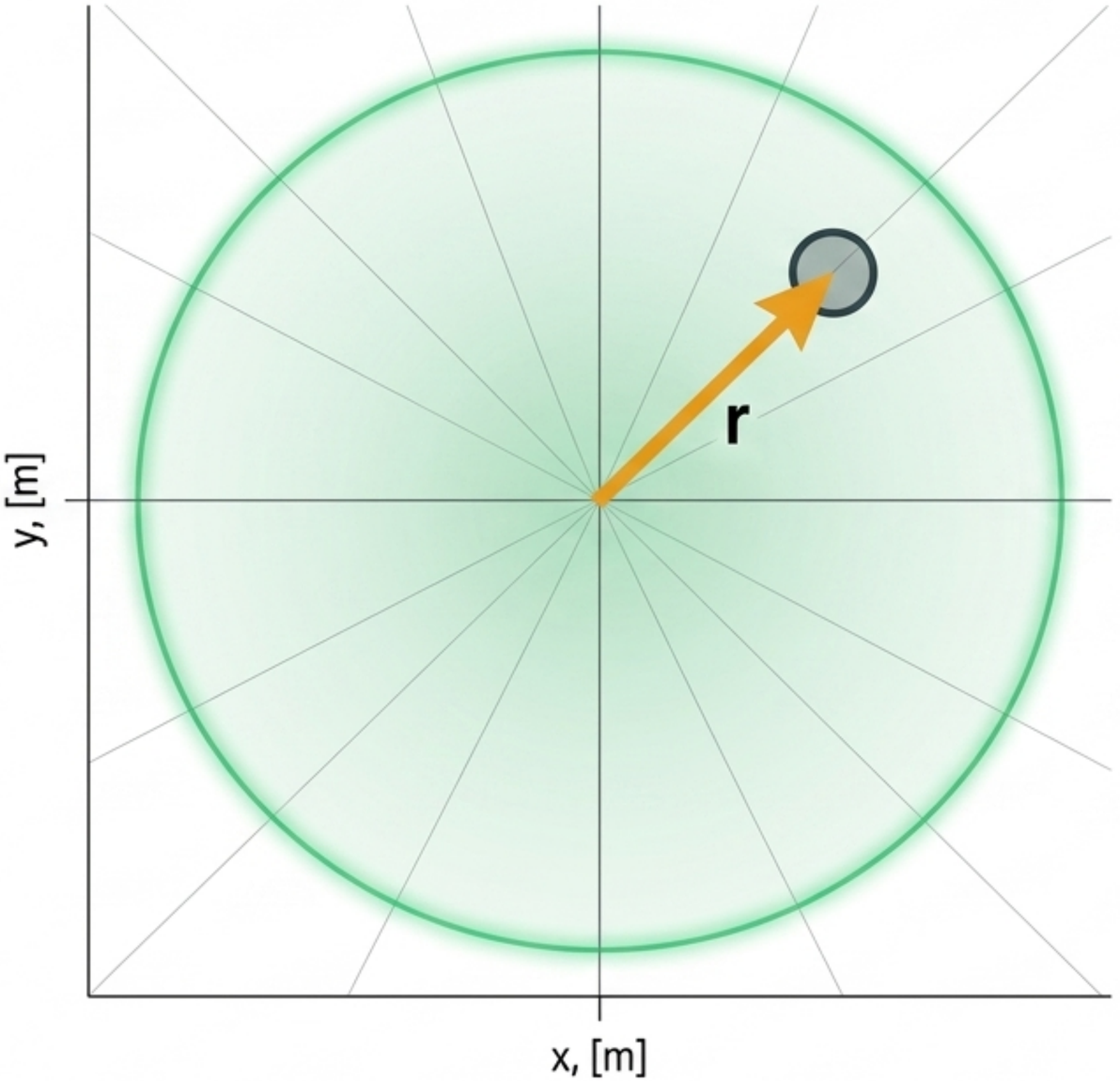


The Physics Reality

Because space is a vacuum, deterministic path loss (h_{PL}) approaches 1. The only factor dictating link survival is the channel random attenuation caused by pointing errors (h_{PE}).

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The Missed Target: Radial Deviation in the Vacuum



The Detector Limit:

The detector radius (w_d) is vastly smaller than the expanded beam ($w_z \gg w_d$) after propagating hundreds of kilometers.

The Pointing Error (r):

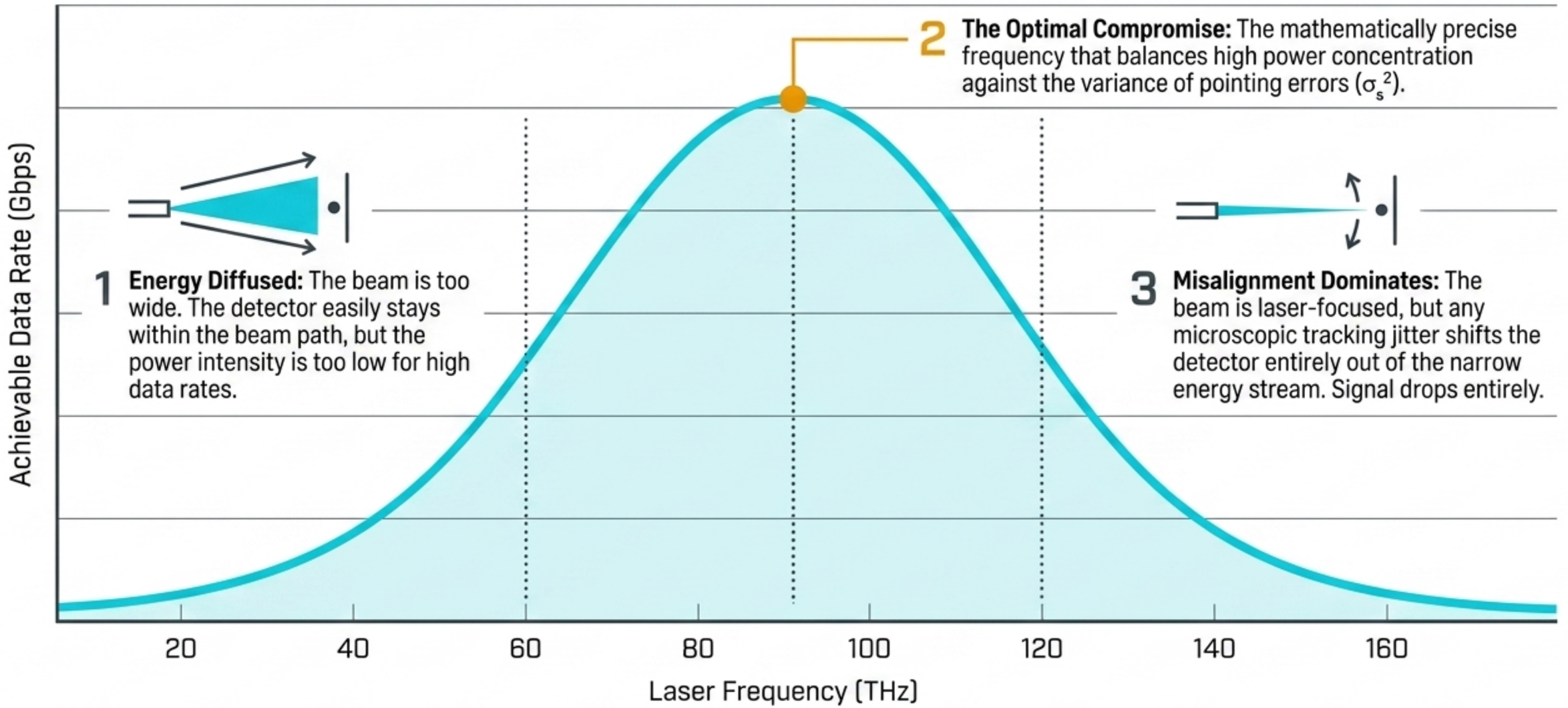
Microscopic satellite jitter causes horizontal and vertical misalignment. This radial deviation follows a Rayleigh distribution.

The Channel State (h_{PE}):

As the detector drifts radially (r) from the beam center, received power intensity plummets exponentially, dropping below the detector's sensitivity threshold (p_{th}).

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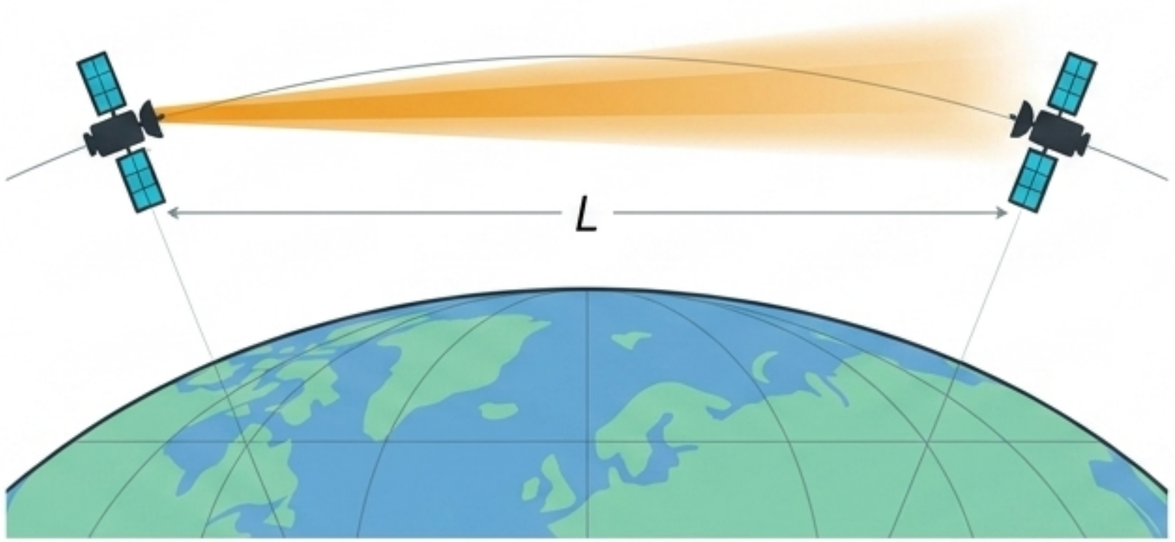
Design Tension 1: Concentration vs. Alignment



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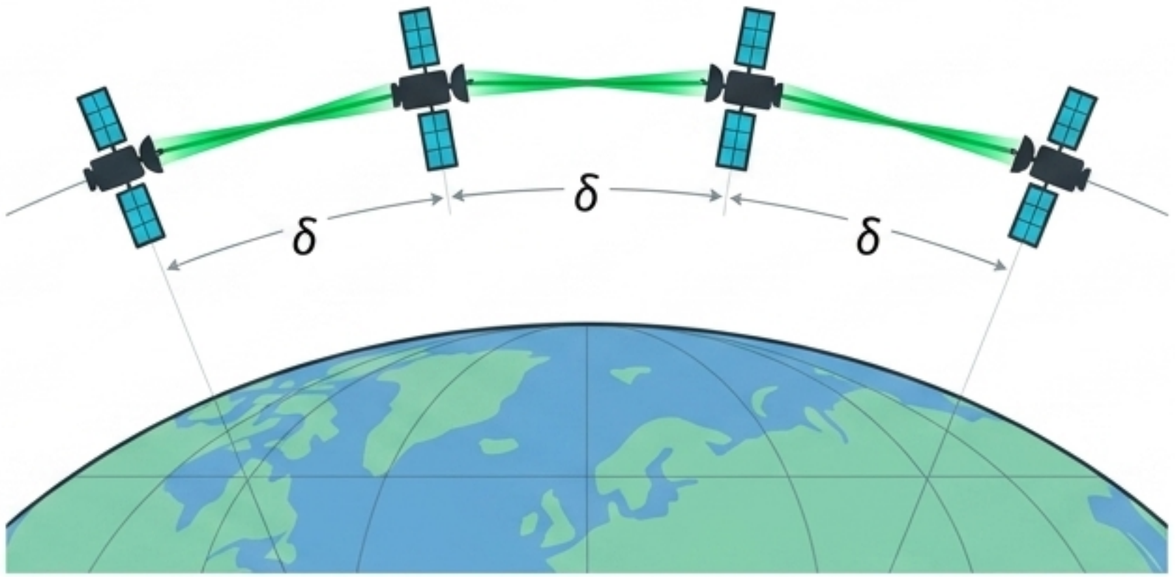
Mitigating Distance Through Cooperative Relays

Single-Hop Failure



Single-Hop Failure: Variance of pointing errors (σ_s^2) scales aggressively with propagation distance (L). A single 3,000 km shot results in massive beam spread, dropping the data rate to near zero.

Multi-Hop Solution



Multi-Hop Solution: Inserting relay satellites divides the massive distance into shorter segments (δ). Shorter links maintain incredibly tight beam waists and minimal radial deviation, drastically increasing the data rate per link (R_n).

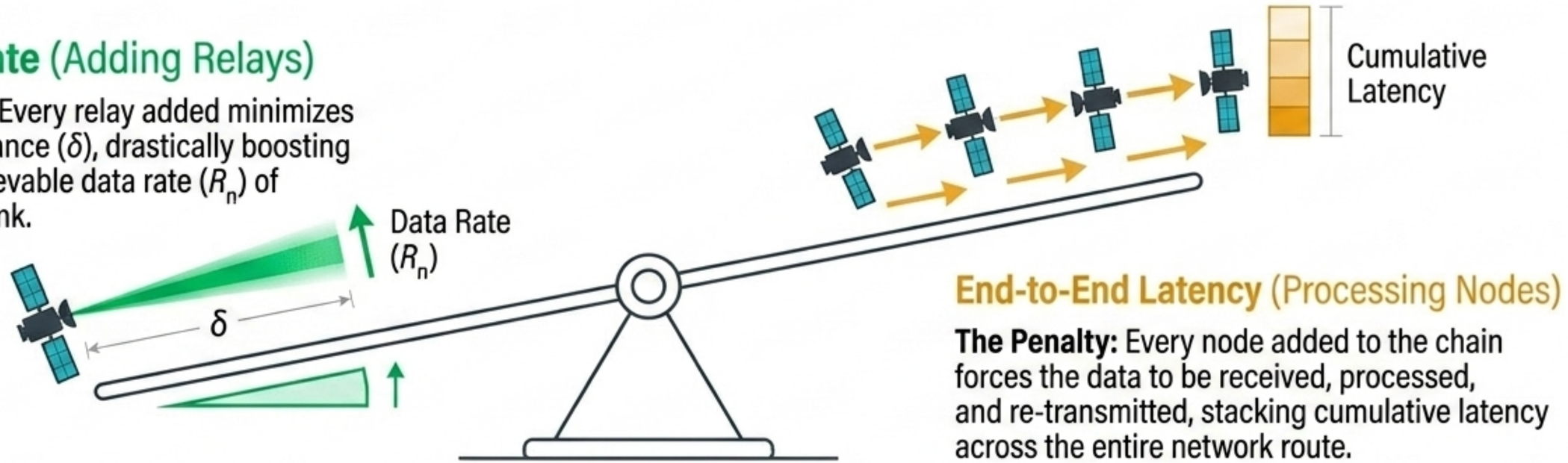
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Design Tension 2: Data Speed vs. Processing Delay

Adding cooperative satellites fundamentally transforms the network equation, introducing a strict mathematical tension:

Link Data Rate (Adding Relays)

The Advantage: Every relay added minimizes propagation distance (δ), drastically boosting the average achievable data rate (R_n) of each individual link.



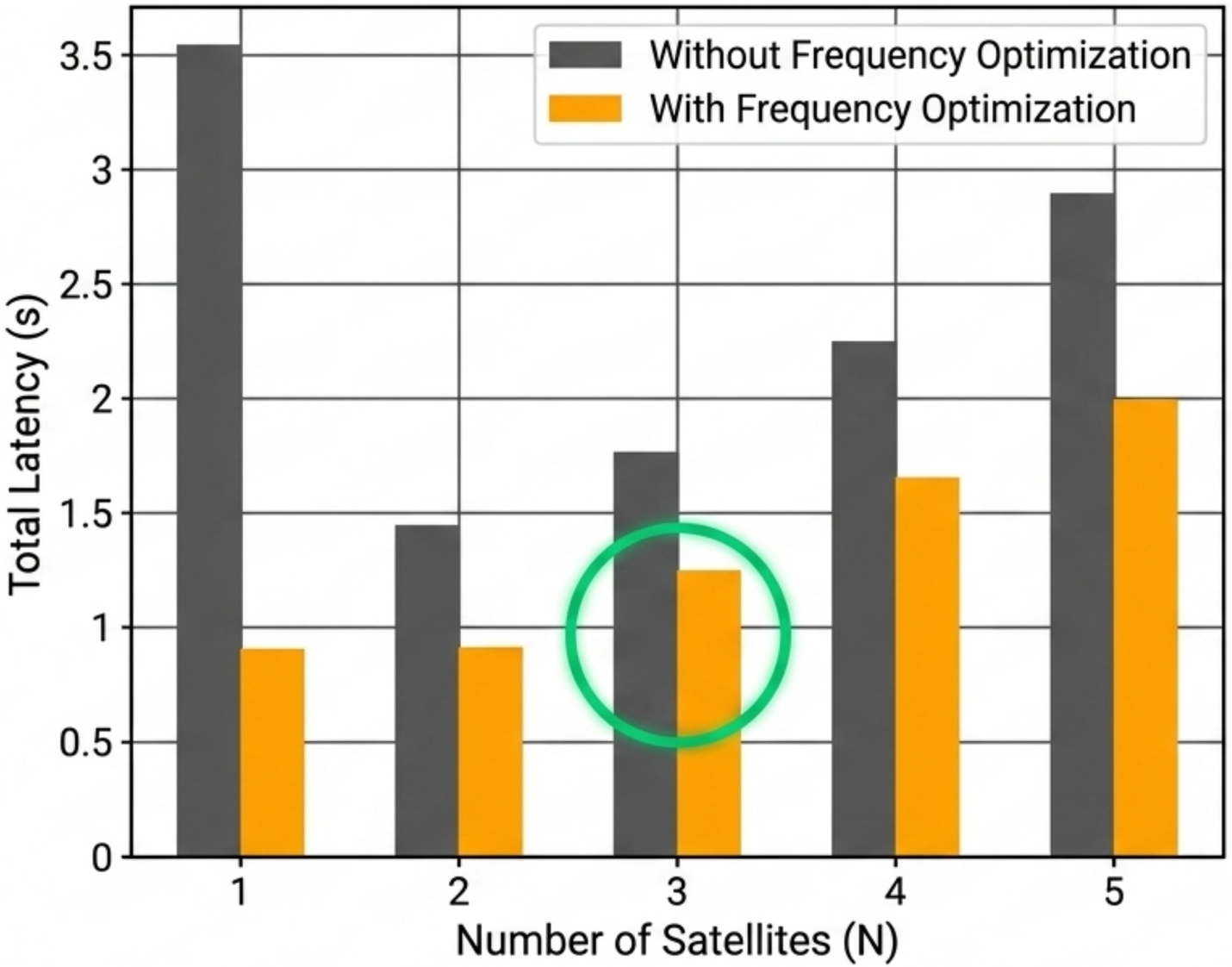
End-to-End Latency (Processing Nodes)

The Penalty: Every node added to the chain forces the data to be received, processed, and re-transmitted, stacking cumulative latency across the entire network route.

The Objective Function: Minimize the number of satellites (N) required to transmit a specific data payload (D) while remaining strictly under a total latency threshold (T_{th}).

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Solving the Optimization Function



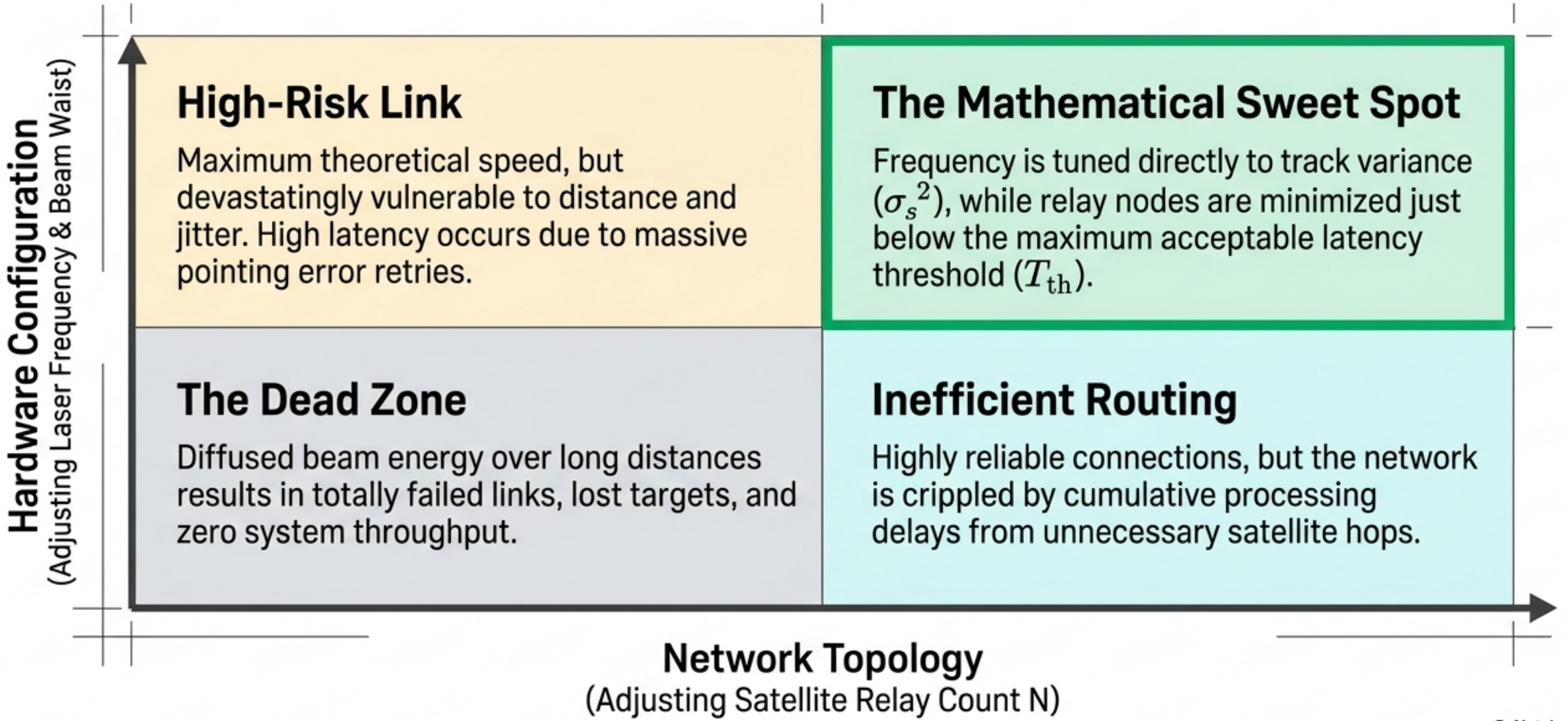
N=1 to N=3: Latency plummets. The raw speed gained from shorter, more accurate multi-hop links heavily outweighs the processing delay of the added satellite nodes.

N=4 to N=5: Latency spikes. The link speeds hit a physical ceiling, and the processing delay of the 4th and 5th hops overwhelms the system efficiency.

The Synergy: At N=3, combining the optimal number of hops with the mathematically optimized laser frequency results in a **65% reduction in total latency** compared to unoptimized baseline links.

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The OISL Architecture Optimization Matrix



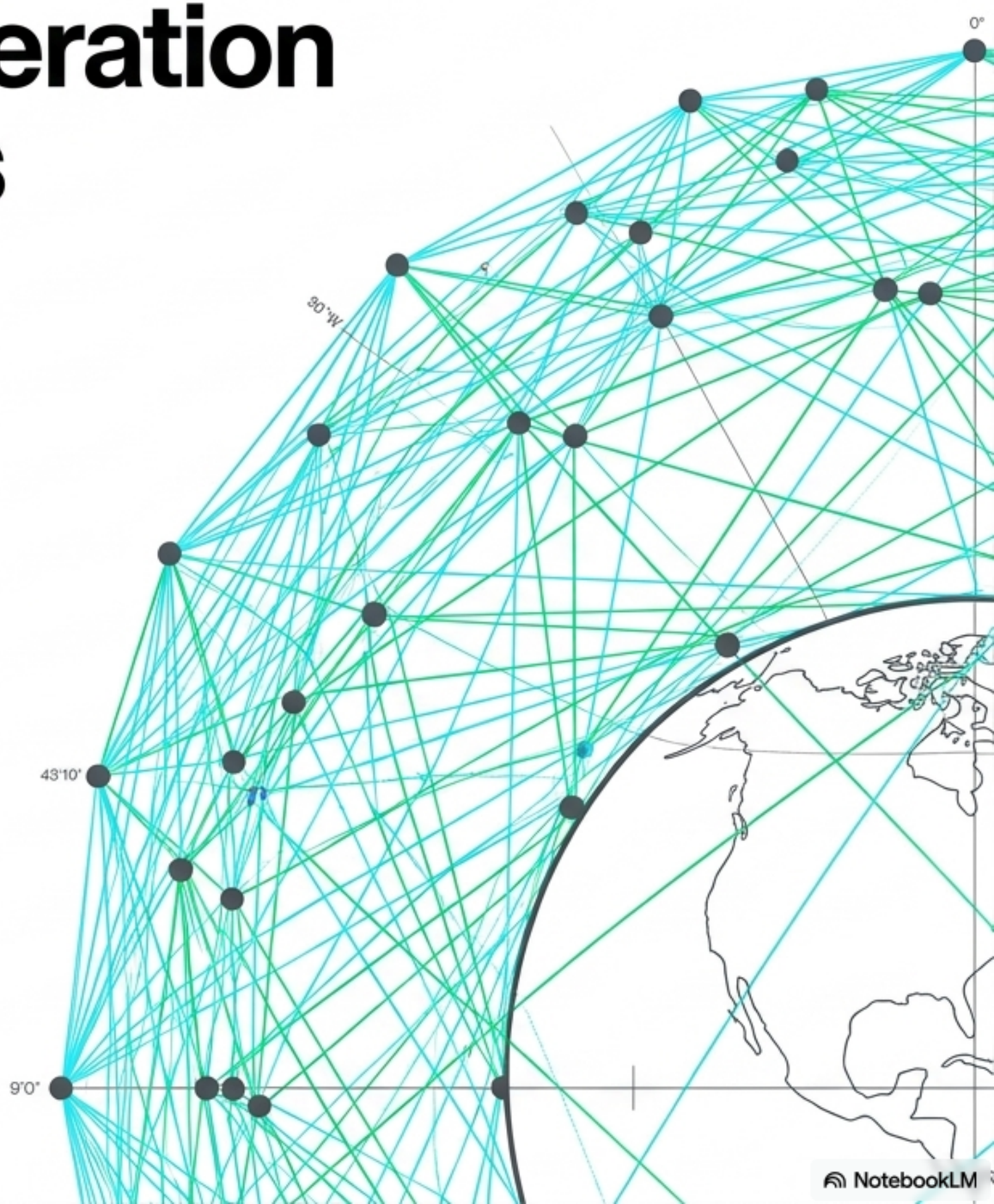
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Powering the Next Generation of Mega-Constellations

As Non-Terrestrial Networks expand, random channel attenuation from pointing errors entirely replaces path loss as the primary engineering constraint in the vacuum of space.

By applying a mathematically rigorous Gaussian beam model, network architects can now dynamically balance physical beam geometry against multi-hop relay topology.

The Result: OISL systems capable of routing high-throughput data across thousands of kilometers with unmatched speed, minimal SWaP footprints, and mathematically optimized latency.



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